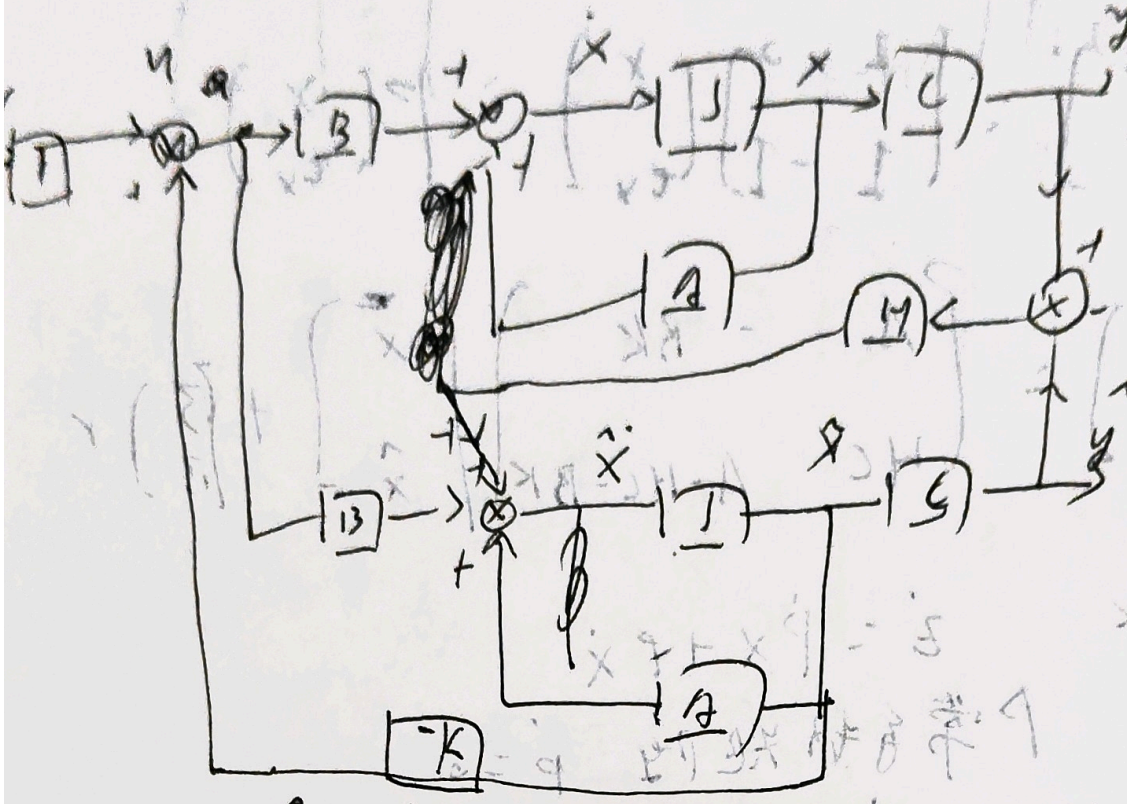
先看课本P94 4.3节



$$\begin{cases} \dot{\hat{x}} = (A - HC)\hat{x} + Bu + Hy \\ \hat{y} = C\hat{x} \end{cases}$$

引入-K, $u \to u = r - K\hat{x}$

~~为什么这作用在 $\hat{y}$ 不在~~ x

$$\dot{x} = Ax + Bu \qquad u = r - K\hat{x}$$

$$= Ax - BK\hat{x} + Br$$

$$\dot{\hat{x}} = \cancel{(A - HC)\hat{x}} + HCx + (A - HC - BK)\hat{x} + Br$$

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ HC & A - HC - BK \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r$$

$$y = [C \quad 0] \begin{bmatrix} x \\ \hat{x} \end{bmatrix} \quad (168)$$

$$e_x = x - \hat{x}$$

$$\begin{bmatrix} x \\ \hat{x} \end{bmatrix} = \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} x \\ e_x \end{bmatrix} = P \begin{bmatrix} x \\ e_x \end{bmatrix}$$

又

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - HC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r$$

$$z = Px \qquad \dot{z} = P\dot{x} + P\dot{x}$$

P 常系数矩阵 $\dot{P} = 0$

$$\dot{z} = P\dot{x}$$

$$\begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{e}_x \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - HC \end{bmatrix} \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} x \\ e_x \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r$$

$$\begin{bmatrix} \dot{x} \\ \dot{e}_x \end{bmatrix} = \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} A - BK & BK \\ 0 & A - HC \end{bmatrix} \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} x \\ e_x \end{bmatrix} + \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} B \\ 0 \end{bmatrix} r$$

分块矩阵求逆，这里不再赘述

$$\begin{bmatrix} I & 0 \\ I & -I \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix}$$

$$\begin{bmatrix} A - BK & BK \\ A - HC & -A + HC \end{bmatrix} \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - HC \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{e}_x \end{bmatrix} = \begin{bmatrix} A-BK & BK \\ 0 & A-HC \end{bmatrix} \begin{bmatrix} x \\ e_x \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} r$$

$$y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ e_x \end{bmatrix}$$

从 $\begin{bmatrix} x \\ \dot{x} \end{bmatrix} \xrightarrow{P} \begin{bmatrix} x \\ xe \end{bmatrix}$ 相似系统变换

具有相同特征多项式

证: $\hat{A} = P^{-1}AP$

$$P_A(\lambda) = \det(\lambda I - A)$$

$$P_{\hat{A}}(\lambda) = \det(\lambda I - P^{-1}AP)$$

$$= \det(P^{-1}(\lambda I)P - P^{-1}AP)$$

$$= \det(P^{-1}(\lambda I - A)P)$$

行列式乘法性质 $\det(XYZ) = \det X \det Y \det Z$

$$\det(P^{-1}) = \det(P)$$

$$\therefore P_{\hat{A}}(\lambda) = P_A(\lambda)$$

$$\begin{bmatrix} A-BK & BK \\ 0 & A-HC \end{bmatrix}$$

求特征多项式

$$\begin{vmatrix} \lambda I-(A-BK) & -BK \\ 0 & \lambda I-(A-HC) \end{vmatrix}$$

$$[\lambda I-(A-BK)][\lambda I-(A-HC)]=0$$

$$\begin{cases} \lambda I-(A-BK)=0 \rightarrow \text{对}(A-BK)\text{矩阵求特征值} \\ \lambda I-(A-HC)=0 \rightarrow \text{对}(A-HC) \end{cases}$$

控制器极点

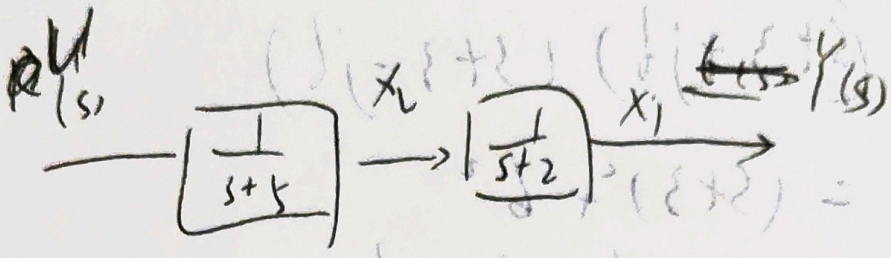
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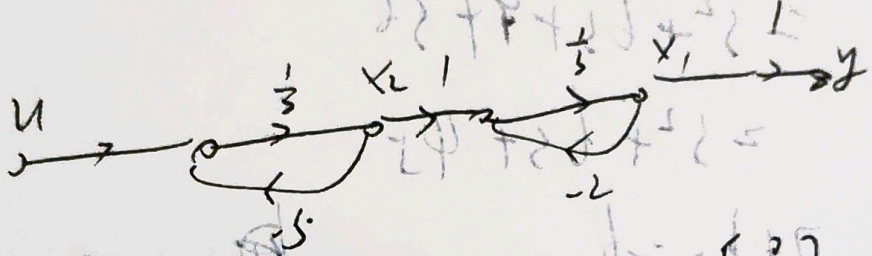
观测器极点

观测器控制器分开设计

例 2A 4.3.2

$$G(s) = \frac{1}{(s+2)(s+5)}$$





$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = Cx = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

满秩 可观

可观测性矩阵 $Q_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

可控性阵 $Q_c = [B, AB] = \begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix}$ 满秩 可控

题目：状态反馈小极点 pole $-3 \pm j6$

观测器 pole $-10, -10$

$$|sI - (A - BK)| \begin{bmatrix} s+2 & -1 \\ k_1 & s+5+k_2 \end{bmatrix} = (s+2)(s+5+k_1)+k_1 = 0$$

由

$$s^2+(2+5+k_2)s+2(5+k_2)+k_1=0$$

$$s^2+(7+k_2)s+(10+k_1+2k_2)=0$$

$$(s+3+j6)(s+3-j6)$$

$$=(s+3)^2+6^2$$

$$=s^2+6s+9+36$$

$$=s^2+6s+45$$

$$7+k_2=6 \qquad k_1=37$$

$$10+k_1+2k_2=45 \qquad k_2=-1$$

A - HC

$$\begin{bmatrix} -2-h_1 & 1 \\ 0-h_2 & -5 \end{bmatrix} \qquad \begin{bmatrix} s+2+h_1 & -1 \\ h_2 & s+5 \end{bmatrix}$$

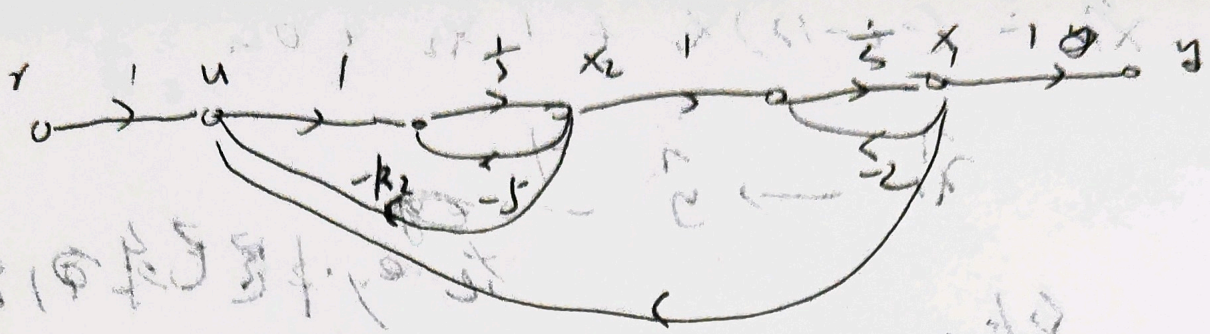
$$(s+2+h_1)(s+5)+h_2=0$$

$$s^2+(7+h_1)s+(2+h_1)\times 5+h_2=0$$

$$s^2+(7+h_1)s+(10+5h_1+h_2)=0$$

$$(s+10)(s+10)=s^2+20s+100$$

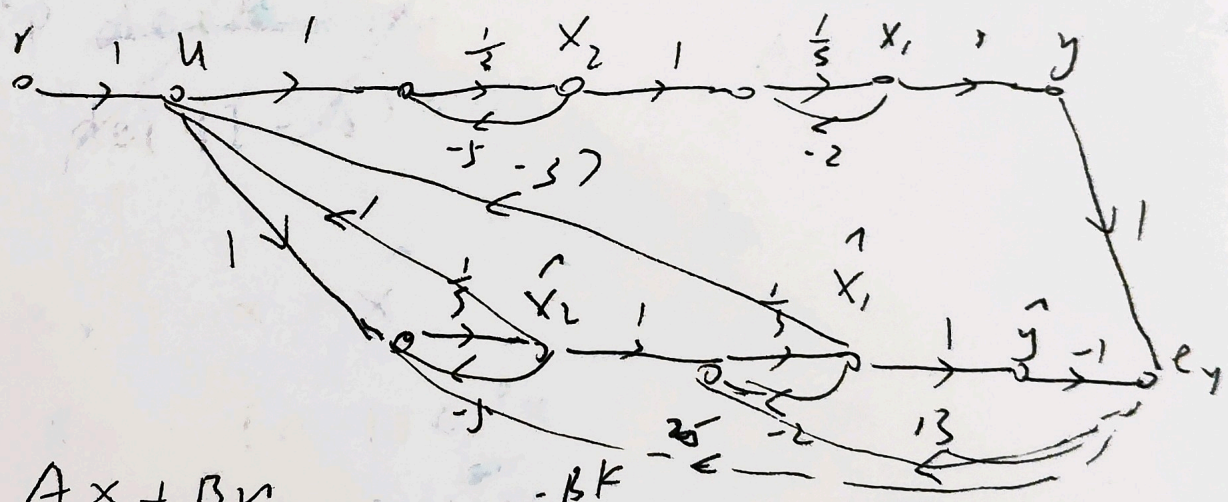
$$\begin{cases} 7+h_1=20 \\ 10+5h_1+h_2=100 \end{cases} \Rightarrow \begin{cases} h_1=13 \\ h_2=25 \end{cases}$$



-K 通过用在 ~~状态变量~~ $\hat{x}$，引回 u

$$u = r - K\hat{x}$$

一条开环



$\dot{x} = Ax + Bu$

$= Ax + B(r - Kx)$

$= (A - BK)x + Br$

$-BK$

$-\begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} k_1 & k_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -k_1 & -k_2 \end{bmatrix}$

$\dot{\hat{x}} = A\hat{x} + Bu + H(y - \hat{y})$

$= (A - HC)\hat{x} + Bu + Hy$

$-\begin{pmatrix} h_1 \\ h_2 \end{pmatrix}\begin{bmatrix} 1 & 0 \end{bmatrix}$

~~$\begin{bmatrix} h_1 & h_2 \end{bmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$~~

$= \begin{bmatrix} -h_1 & 0 \\ -h_2 & 0 \end{bmatrix}$

$\dot{x}_1 = (-2-1)x_1 + 1 \cdot x_2 + 0 \cdot u$

$x_1 \longrightarrow y \longrightarrow$

在 e_y 中是已知的 $y(-1)$

所以从 e_y 引回来直接是 $h_1 = 1$

但从 $\hat{x}_1$ 引回 h_1 是 $-k_1$

